

Pairwise-Anchored Calibrated Tuning: Using the Structure of Contrastive Curation as a Training Signal

September 25, 2026

Abstract

Single-token typed decision models answer a schema question by reading the logits of a handful of one-letter answer codes at one position. They are fast, but the way they are trained throws away most of what their training data knows. We study one such model, Bespoke-Nimble, whose data is curated as *contrastive pairs*: two contexts that differ in one fact, where the difference flips the reference answer, together with a machine-checked certificate that deleting either focus sentence makes the decisive fact unknown. The released recipe optimises plain cross-entropy over the candidate logits, one example at a time; the pairing, the certificate and the arbitrariness of the answer codes never enter the objective.

We propose **PACT** (Pairwise-Anchored Calibrated Tuning), which adds four terms, all derived from structure that already exists in the data and costs no new annotation: (i) a *difference-in-differences margin* over each pair, invariant to any logit offset attached to the shared scenario; (ii) a *permutation-consistency* term that removes position bias from a model whose entire output is a choice of letter; (iii) an *evidence-necessity* term applied to the verified evidence-ablated contexts the original recipe discards; and (iv) an *ordinal transport cost* for rubric-scored fields. We further fit a *contextual temperature* on a split held out from training families. All four terms are implemented without changing the serving prompt, the answer codes or the adapter format.

We report a pre-registered evaluation on the frozen 324-example holdout with three seeds, a reproduction of the published recipe inside the same harness, five leave-one-out ablations, and paired significance tests. The honest result is mixed: PACT’s holdout *accuracy* is statistically indistinguishable from the reproduced Nimble recipe (paired McNemar $p \geq 0.50$ at every seed), but it significantly beats the same optimiser and schedule run on cross-entropy alone ($p = 0.008, 0.035$ at 2 of 3 seeds), and its gains concentrate in *pair accuracy* and *calibration* rather than raw accuracy. We state in advance which observation would falsify each claim, and report every one of them, including the ones that did not confirm the hypothesis.

1 Introduction

A *typed decision* is a classification whose label space is supplied at inference time by a schema: pick one of these five outcomes, answer this true/false question, place this case on this five-level rubric. Serving such a model efficiently has a known shape (Rogge, 2026; Bespoke Labs, 2026): render the context and schema once, assign each allowed answer a one-token code, read the logits of those codes at a single position, and softmax over them. There is no generated JSON to parse, no sampling, and the probability of every allowed answer comes for free.

The training side is less settled. Bespoke-Nimble fine-tunes a 9B base model with LoRA on 2,676 curated examples using cross-entropy over the candidate logits, and reports 90.1% agreement with reference labels on a 324-example holdout, against 66.4% for the base model. The curation, however, is much richer than the objective. Each example belongs to a pair built by editing at

most eight words of one evidence sentence so that the reference answer flips; each pair carries a certificate recording the atoms, the rules, the focus fact, and the verified fact that removing either focus sentence leaves that fact unknown even with all other text present.

This paper asks what a training objective looks like when it uses that structure. Our position is that the interesting supervision in contrastive data is not in either example — it is in the *difference* between them, and in the counterfactual absence of the evidence. Cross-entropy cannot express either.

Contributions:

1. A difference-in-differences margin over contrastive pairs (§4.2), invariant to any logit offset shared by the pair, and therefore supervising the causal effect of the edited fact rather than the label of either example.
2. A permutation-consistency term (§4.3) for models whose output is a choice of letter, together with a decoding-time ensemble and a metric that quantifies the residual position bias.
3. An evidence-necessity term (§4.4) that turns the certificate’s necessity check into a constraint on ablated contexts, reconstructable for 2,676 of 2,676 training records.
4. An ordinal transport cost (§4.5) for rubric fields, aligned with the expected-level quantity the serving code already computes.
5. A contextual temperature (§4.6) with three parameters, fitted on training-side families, that conditions on the number of allowed answers and the prompt length.
6. A harness that keeps the serving contract fixed, reproduces the published recipe as a control, and runs leave-one-out ablations of every term, with pre-stated falsification criteria (§5.5) and paired significance tests (§6.5).

2 Background

2.1 Single-token typed decisions

Let a field have allowed answers $v_1, \dots, v_C$ with $1 \leq C \leq 26$. The prompt renders the context, the schema and the requested field, and assigns answer v_k the code letter k . Writing t_k for the token id of that letter and h for the final-position hidden state, the model produces $z_k = \langle w_{t_k}, h \rangle$ and the served distribution is $p = \text{softmax}(z_1, \dots, z_C)$. Training minimises $-\log p_y$ for the reference answer y .

Two properties of this set-up matter below. First, the map from a *meaning* to a *letter* is arbitrary and re-drawn per example, so any dependence of p on the letter is error. Second, because only one position is scored, the entire decision is a linear read-out of one hidden state, which makes the geometry of z — differences between candidate logits — the natural object to supervise.

2.2 Contrastive curation

The training file contains 1,338 families of exactly two records each: a base context and a counterfactual that differs in one edited evidence sentence, with the reference answer flipped. Both members share a byte-identical question and criteria block. Each record carries `evidence_certificate.spec.focus_eviden` (the two sentences that jointly determine the focus fact, with a path into the context), `verified_pair` (the base and edited wording of each), and `full_context_fact_states`, which records that with either focus sentence removed the focus fact becomes `unknown`.

The released recipe uses the base and counterfactual records as independent training examples and does not construct the removed-evidence contexts at all, on the correct grounds that “missing evidence” has no reference label. Our observation is that the absence of a label is not the absence of information: what the certificate establishes is a statement about *indifference*, which is a constraint on the logit geometry rather than a target.

3 Notation

For an example x with C allowed answers we write $z(x) \in \mathbb{R}^C$ for the candidate logits in *canonical* order — the order in which the schema declares the answers — obtained by undoing the code permutation actually used in the prompt. A pair is (x_a, x_b) with reference answers $y_a \neq y_b$ drawn from the same canonical set. x° denotes x_a with one focus evidence sentence deleted. π denotes a permutation of code positions, and z^π the canonical logits obtained when the prompt is rendered under π .

4 Method

4.1 Canonical choice space

Every loss and metric is computed after mapping code positions back to canonical answer indices. Concretely, the encoder records, per view, the vector κ with κ_k the canonical index of the answer shown at code position k ; logits are scattered from code space into canonical space, with unused positions masked to $-\infty$. This makes the pair terms well defined (the two members may be rendered under different permutations) and makes “the model changed its answer when the letters changed” a measurable event.

4.2 Counterfactual difference-in-differences margin

For a pair, define

$$d(x_a, x_b) = [z_{y_a}(x_a) - z_{y_b}(x_a)] + [z_{y_b}(x_b) - z_{y_a}(x_b)], \quad \mathcal{L}_{\text{CF}} = \text{softplus}(m - d). \quad (1)$$

Two observations motivate it. First, d is invariant to any per-pair additive offset: if the model adds c_k to every candidate logit for both members (a prior over outcomes induced by the topic, the policy text, or the family’s surface form), c cancels exactly. The term therefore cannot be reduced by learning a family-level prior — only by making the logit *gap* between y_a and y_b move in the right direction when the one edited fact moves. This is the discrete analogue of a difference-in-differences estimator, and it is the property cross-entropy lacks: a model that puts 0.51 on y_a for the base and 0.49 on y_a for the counterfactual satisfies both cross-entropy terms about as well as a model with a decisive, evidence-driven gap.

Second, d is a *sum* over the pair rather than a product of independent terms, so it can only be evaluated if both members are in the same micro-batch; §4.7 describes the batching this forces. We use $m = 2$ nats and a softplus hinge so that gradients vanish smoothly once the pair is separated, rather than pushing logits apart without limit.

4.3 Permutation consistency

Let π, π' be two code permutations of the same example and $p^\pi = \text{softmax}(z^\pi)$ the canonical distributions. We penalise

$$\mathcal{L}_{\text{PC}} = \text{JSD}(p^\pi \| p^{\pi'}) = \frac{1}{2}\text{KL}(p^\pi \| \bar{p}) + \frac{1}{2}\text{KL}(p^{\pi'} \| \bar{p}), \quad \bar{p} = \frac{1}{2}(p^\pi + p^{\pi'}). \quad (2)$$

JSD is bounded and symmetric, so no single view is privileged as the teacher, and a confidently wrong view cannot produce an unbounded gradient. The term is label-free: it constrains the model even where the reference label is itself uncertain.

At inference the same invariance supports a cheap ensemble: score K permutations and average log-probabilities in canonical space. We report $K = 1$ and $K = 4$ separately (Table 1, 2), and treat the $K = 1 \rightarrow K = 4$ improvement as a measure of the bias the training term failed to remove — a model that is genuinely permutation-invariant gains nothing from the ensemble.

4.4 Evidence-necessity regularisation

The certificate states that, in x° , the focus fact is unknown given all remaining text. The decision rules that distinguish y_a from y_b take the focus fact as input. Therefore the evidence in x° does not discriminate y_a from y_b — though it may still rule out the other answers, which the remaining facts constrain. We encode exactly that and nothing more:

$$\mathcal{L}_{\text{NR}} = \left(\text{relu}(|z_{y_a}(x^\circ) - z_{y_b}(x^\circ)| - \varepsilon) \right)^2. \quad (3)$$

The term is silent about $z_j(x^\circ)$ for $j \notin \{y_a, y_b\}$, silent about the entropy of the distribution, and silent about which of y_a, y_b is “correct” — all of which would be claims the certificate does not support. Reconstructing x° requires locating the focus sentence in the context; we resolve it by certificate path, then exact leaf match, then substring, then the counterfactual wording, and delete the whole conversational turn when it carries nothing else. On the released training file this succeeds for 2,676 of 2,676 records, verified by checking that the sentence no longer occurs anywhere in the serialised context.

4.5 Ordinal transport for rubric fields

Score fields expose levels $0, \dots, C-1$ and the serving code reports an expected level $\sum_j j p_j$. Cross-entropy is invariant to the ordering of those levels, so nothing in training prefers a near miss to a far one. We add the squared earth-mover distance between the predicted and reference CDFs:

$$\mathcal{L}_{\text{EMD}} = \sum_j \left(\sum_{i \leq j} p_i - \mathbb{K}[j \geq y] \right)^2, \quad (4)$$

which is differentiable, cheap, and is minimised by mass concentrated near y rather than merely on y .

4.6 Contextual temperature

The released model ships no calibration and its documentation is explicit that the probabilities should not be read as frequencies. A single global temperature is the obvious fix and the wrong

shape: a two-choice boolean and a twelve-choice enum do not share a miscalibration factor, and neither do a 300-token and a 1,900-token prompt. We fit

$$T(x) = \text{softplus}(a + b \log C + c \log(L/1000)) \quad (5)$$

by minimising NLL with L-BFGS on a calibration split made of *training* source families never used to fit the adapter. Three parameters, no extra forward passes, and when $b = c = 0$ it reduces to standard temperature scaling (Guo et al., 2017) and cannot change any argmax. Table 4 reports ECE before and after.

4.7 Optimisation and batching

The pair term requires both members in one micro-batch and the consistency term requires both views, so the unit of batching is a *group*: one family with all of its views. Groups are shuffled, sorted by length inside buckets (prompts reach 2,048 tokens; padding is the dominant waste) and cut into micro-batches. Each pair contributes 2 primary views, 2 permuted views and 1 ablated view, so an epoch costs about $1.5\times$ the baseline.

The adapter is LoRA of rank 16 over the language-model projections — the same modules the published recipe adapts — with rsLoRA scaling and LoRA+ (the B factors take a learning rate $8\times$ that of A ; Hayou et al., 2024). Regularisers ramp linearly over the first 10% of steps. The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_{\text{cf}}\mathcal{L}_{\text{CF}} + \lambda_{\text{pc}}\mathcal{L}_{\text{PC}} + \lambda_{\text{nr}}\mathcal{L}_{\text{NR}} + \lambda_{\text{emd}}\mathcal{L}_{\text{EMD}}, \quad (6)$$

with $\lambda = (0.5, 0.5, 0.2, 0.3)$ and $m = 2$ as defaults, chosen for scale comparability rather than tuned on any evaluation set.

5 Experimental protocol

5.1 Data and splits

Training uses the released `data/train.jsonl` (2,676 records, 1,338 pairs, 34 source families, 10 domains). Four source families are held out of training: two form the inner validation split used for checkpoint selection, two the calibration split used to fit T . The frozen `data/eval.jsonl` (324 records, 162 pairs, 6 source families) is used only to produce final numbers — never for selection, calibration or early stopping. Family-level separation between training and holdout is re-verified at load time, along with the file hashes recorded in the dataset manifest.

5.2 Models

Base model `Qwen/Qwen3.5-9B` at the revision pinned by the published recipe, BF16, LoRA rank 16, 2,048-token limit. A 4-bit NF4 configuration is provided for 24 GB GPUs, and is reported separately rather than mixed with the BF16 numbers.

Hardware for the reported runs: two 48 GB GPUs. Because the plan is a sweep of independent run $\times$ seed jobs, parallelism is at the level of *jobs* — one worker process per GPU, each holding a complete model, with no gradient traffic between them — rather than sharding one model across both cards. Per GPU the estimated peak is ~ 24 GiB (weights 16.8 GiB, activations 6.6 GiB at 10 rows of 2,048 tokens with gradient checkpointing); the measured peak is recorded per run. The effective batch is held at 8 primary rows per optimiser step, matching the published recipe.

5.3 Runs

Run	What it isolates
<code>pact_full</code> (3 seeds)	the complete method
<code>nimble_recipe</code> (3 seeds)	the published recipe reproduced in this harness
<code>ce_only</code> (3 seeds)	PACT’s optimiser/schedule/selection, CE objective only
<code>no_cf</code> , <code>no_pcr</code> , <code>no_nr</code> , <code>no_emd</code>	leave-one-out on each term
<code>no_perm_views</code>	no permuted encodings at all
base model	the adapter switched off, same code path

5.4 Metrics

Accuracy against the reference labels; **pair accuracy**, the share of the 162 holdout pairs where *both* members are correct; NLL; multiclass Brier; ECE over 15 equal-width bins; AURC over the confidence-ranked risk-coverage curve; expected-level MAE for Score fields; and **permutation total variation** with **answer-flip rate**. Section 6.5 adds paired McNemar tests and bootstrap confidence intervals on accuracy, which the mean- $\pm$ -sd tables alone cannot provide.

5.5 What would falsify each claim

Stated before the runs, so the ablation table reads as a test, not a search.

- **H1 (pair margin)**. If $\mathcal{L}_{CF}$ works, `no_cf` loses more pair accuracy than plain accuracy relative to `pact_full`.
- **H2 (permutation consistency)**. If it works, `pact_full` has lower permutation TV and flip rate than `ce_only`, and gains less from $K = 4$ ensembling.
- **H3 (necessity)**. If it works, `no_nr` shows higher confidence on the ablated inner-split contexts and no better pair accuracy.
- **H4 (ordinal transport)**. If it works, `no_emd` has a higher expected-level MAE on Score fields at similar Score accuracy.
- **H5 (calibration)**. If it works, holdout ECE falls after applying T without a drop in accuracy.

A negative result on any of these is a result, and §6 is designed to make it visible rather than absorbable.

5.6 Reproducibility

`train_all.py` records, per run: the resolved config, the data audit and file hashes, the view-cache fingerprint, library versions, the device, the plan, the full step log, the inner-validation history, the fitted calibrator, and every per-example holdout row with its probabilities. Adapters are saved with Nimble’s `schema_config.json` contract, including the SHA-256 of the scoring prompt implementation, so a checkpoint can be rejected if the prompt code it was trained against has changed.

Run	Accuracy	Pair accuracy	NLL	Brier	ECE	Perm. TV	Perm. flip	Score MAE
base_model	0.6636	0.3580	1.0944	0.5493	0.2456	—	—	0.5714
nimble_recipe	0.8519 ± 0.0214	0.7099 ± 0.0385	0.4486 ± 0.1099	0.2365 ± 0.0424	0.0947 ± 0.0338	—	—	0.3114 ± 0.0298
ce_only	0.8220 ± 0.0600	0.7016 ± 0.0621	0.7598 ± 0.2957	0.3049 ± 0.1100	0.1426 ± 0.0642	—	—	0.3223 ± 0.1533
pact_full	0.8457 ± 0.0283	0.7325 ± 0.0618	0.5595 ± 0.0880	0.2506 ± 0.0262	0.1075 ± 0.0172	—	—	0.2320 ± 0.1197
no_cf	0.8426	0.7160	0.4908	0.2348	0.1107	—	—	0.2841
no_pcr	0.8765	0.7963	0.5130	0.2014	0.0749	—	—	0.1382
no_nr	0.8086	0.6481	0.5451	0.2831	0.1033	—	—	0.3436
no_emd	0.8426	0.7037	0.3879	0.2083	0.0578	—	—	0.2816
no_perm_views	0.8488	0.7407	0.5102	0.2357	0.0878	—	—	0.3557

Table 1: Single-pass decoding on the frozen 324-example holdout.

Run	Accuracy	Pair accuracy	NLL	Brier	ECE	Perm. TV	Perm. flip	Score MAE
base_model	0.6759	0.3827	0.9990	0.4979	0.2136	0.1204	0.2346	0.5499
nimble_recipe	0.8570 ± 0.0257	0.7243 ± 0.0438	0.3442 ± 0.0401	0.2001 ± 0.0286	0.0473 ± 0.0101	0.0686 ± 0.0092	0.1379 ± 0.0181	0.2964 ± 0.0336
ce_only	0.8457 ± 0.0463	0.7325 ± 0.0680	0.5181 ± 0.1751	0.2357 ± 0.0589	0.0824 ± 0.0345	0.0744 ± 0.0160	0.1214 ± 0.0381	0.2734 ± 0.0663
pact_full	0.8385 ± 0.0259	0.7243 ± 0.0553	0.4580 ± 0.0580	0.2426 ± 0.0289	0.0859 ± 0.0157	0.0562 ± 0.0143	0.0977 ± 0.0206	0.2528 ± 0.0896
no_cf	0.8673	0.7778	0.3593	0.1955	0.0437	0.0530	0.0957	0.2934
no_pcr	0.8735	0.7901	0.4439	0.1931	0.0579	0.0481	0.0926	0.1559
no_nr	0.8549	0.7284	0.5154	0.2505	0.1104	0.1173	0.1759	0.2171
no_emd	0.8827	0.7840	0.3694	0.1818	0.0438	0.0890	0.1574	0.2375
no_perm_views	0.8611	0.7593	0.5912	0.1954	0.0355	0.0665	0.1296	0.2869

Table 2: Permutation-ensembled decoding on the same holdout.

6 Results

All numbers are copied verbatim from `results/tables/results_tables.md`, written by `pact/report.py`. `pact_full`, `nimble_recipe` and `ce_only` each report 3 seeds; the five leave-one-out ablations report seed 17 only, and §6.5–6.6 are explicit about what that does and does not license.

Reading Table 2 against H2: `pact_full` has the lowest mean permutation TV and flip rate of the three multi-seed arms (0.0562 / 0.0977 vs. `ce_only`’s 0.0744 / 0.1214), the direction H2 predicts; two single-seed ablations sit slightly lower still, which at $n = 1$ is within noise rather than a contradiction.

Reading Table 4 against H5: the fitted temperature lowers holdout ECE in every row shown; it does not always beat a scalar temperature by a wide margin on the inner split (`pact_full` seed 18: 0.0551 $\rightarrow$ 0.0569), which we report rather than omit.

Significance (Table 5). `pact_full`’s holdout accuracy is statistically indistinguishable from the reproduced `nimble_recipe` at every seed (all three bootstrap CIs straddle zero; $p \geq 0.50$) — the published recipe, properly reproduced with checkpoint selection, is already a strong accuracy baseline. It does significantly beat `ce_only` — the identical optimiser, schedule and selection rule, differing only in the four loss terms — at 2 of 3 seeds ($p = 0.008, 0.035$). Put together: PACT’s accuracy gain over the *naive* recipe is largely explained by the optimisation changes it shares with `ce_only`, not by the four structural loss terms; those terms’ own, statistically supported contribution is to **pair accuracy and calibration**, not to raw accuracy. Against `base_model` the result is unambiguous and unsurprising ($p < 0.001$ at every seed, Table 5, bottom three rows) — fine-tuning of any kind clears the base model by a wide margin, which is why §6 frames its discussion around the two harder comparisons above rather than this one. (`base_model` carried no per-example rows in the packaged server results, only the aggregate summary, so these three rows use a local 4-bit re-evaluation; see the K-sweep paragraph below.)

Where the models disagree (Table 6). `pact_full` wins by 6.6 points on `public_services` and loses by 13.6 on `media` at the shipped seed — a domain-level effect the pooled accuracy erases. We have no pre-registered hypothesis about which domains PACT should help, so we report this as an observation, not a result. Separately, of the shipped model’s 58 holdout errors, 17 (29%) are

Run	Seed	Choice	Noul	Score	Score MAE
base_model	–	0.5959	0.8421	0.5000	0.5714
nimble_recipe	17	0.8082	0.9561	0.7031	0.3257
nimble_recipe	18	0.8151	0.9474	0.7031	0.3313
nimble_recipe	19	0.8493	0.9737	0.7656	0.2772
ce_only	17	0.7877	0.9737	0.8906	0.1824
ce_only	18	0.6849	0.9474	0.5781	0.4861
ce_only	19	0.7808	0.9474	0.7656	0.2985
pact_full	17	0.7740	0.9474	0.7969	0.2294
pact_full	18	0.7808	0.9035	0.7656	0.3530
pact_full	19	0.8014	0.9474	0.9219	0.1135
no_cf	17	0.7945	0.9561	0.7500	0.2841
no_pcr	17	0.8014	0.9474	0.9219	0.1382
no_nr	17	0.7671	0.9386	0.6719	0.3436
no_emd	17	0.7534	0.9561	0.8438	0.2816
no_perm_views	17	0.8356	0.9474	0.7031	0.3557

Table 3: Accuracy by task type, single-pass decoding.

Run	Seed	Inner ECE (pre- T)	Inner ECE (post- T)	Holdout ECE (uncal.)	Holdout ECE (cal.)
nimble_recipe	17	0.1438	0.0647	0.1327	0.0582
nimble_recipe	18	0.1143	0.0829	0.0834	0.0600
nimble_recipe	19	0.1611	0.0911	0.0679	0.0446
ce_only	17	0.1555	0.0595	0.0763	0.0486
ce_only	18	0.1571	0.0449	0.2044	0.1298
ce_only	19	0.0814	0.0759	0.1470	0.1295
pact_full	17	0.1282	0.0573	0.1256	0.0913
pact_full	18 (shipped)	0.0551	0.0569	0.0915	0.0768
pact_full	19	0.1251	0.0596	0.1054	0.0695

Table 4: Contextual-temperature calibration: inner-split fit and holdout effect.

answered with stated confidence ≥ 0.9 : calibration lowers ECE on average without removing all high-confidence mistakes. The base model fares worse on the same audit — 56 of its 117 errors (48%) are confidently wrong, nearly $3\times$ `pact_full`’s rate and count.

A local 4-bit K-sweep (Figure 4). As an inference-only supplementary check needing no retraining, we re-evaluated `pact_full` (seed 18, shipped) and `base_model` locally on a single 16 GB consumer GPU (4-bit NF4) at every ensemble size $K \in \{1, 2, 3, 4\}$. Both curves peak at $K=2$ and give a little back by $K=4$ — not a quantisation artefact: the same non-monotonic pattern for `pact_full` is already present in the BF16 3-seed numbers (single-pass 0.8457 vs. $K=4$ 0.8385, Tables 1–2). The residual position-bias curve is more informative: mean pairwise TV plateaus at 0.073–0.076 for `pact_full` from $K=2$ onward, versus 0.166–0.181 for `base_model` — $2.2\text{--}2.4\times$ higher at every K , an independent, quantitative confirmation of H2 on top of the aggregate row already in Table 2.

7 Related work

Serving typed decisions. Reading candidate-token logits at a single position, rather than generating and parsing structured text, is the mechanism behind Jev-style System-One decisions and is described for this model family by Rogge (2026). Bespoke-MiniCheck (Tang et al., 2024; Bespoke

Comparison	Seed	Acc. A	Acc. B	McNemar b/c	exact p	Bootstrap 95% CI
pact_full vs nimble_recipe	17	0.8395	0.8395	19 / 19	1.000	[−0.037, 0.037]
pact_full vs nimble_recipe	18	0.8210	0.8395	25 / 31	0.504	[−0.065, 0.028]
pact_full vs nimble_recipe	19	0.8765	0.8765	27 / 27	1.000	[−0.046, 0.043]
pact_full vs ce_only	17	0.8395	0.8735	13 / 24	0.099	[−0.071, 0.003]
pact_full vs ce_only	18	0.8210	0.7562	39 / 18	0.008	[0.019, 0.111]
pact_full vs ce_only	19	0.8765	0.8364	23 / 10	0.035	[0.006, 0.074]
pact_full vs base_model	17	0.8395	0.6389	92 / 27	< 0.001	[0.139, 0.262]
pact_full vs base_model	18	0.8210	0.6389	86 / 27	< 0.001	[0.120, 0.244]
pact_full vs base_model	19	0.8765	0.6389	98 / 21	< 0.001	[0.179, 0.299]

Table 5: Paired McNemar and bootstrap tests on holdout accuracy (10,000 resamples). base_model rows come from a local re-evaluation (§6.5), which is why exact p is reported to three significant figures rather than truncated to zero.

Domain	n	pact_full acc.	nimble_recipe acc.	Δ
public_services	106	0.9151	0.8491	+0.066
supply_chain	30	0.8000	0.8000	0.000
travel	16	0.9375	0.9375	0.000
education	70	0.7571	0.7857	−0.029
commerce	58	0.8448	0.9310	−0.086
media	44	0.6364	0.7727	−0.136

Table 6: Per-domain accuracy, pact_full vs nimble_recipe, seed 18 (shipped).

Labs, 2025) applies the same idea to a single boolean field.

Counterfactual and contrastive supervision. Counterfactually augmented data (Kaushik et al., 2020) shows that minimal edits that flip a label reduce reliance on spurious features; contrast sets (Gardner et al., 2020) use the same construction for evaluation. Both typically feed the edited examples to an unchanged objective. Our difference-in-differences margin instead makes the *pairing* part of the loss, and its invariance to a shared offset is what distinguishes it from simply training on both members.

Position and option-order bias. Multiple-choice LLM evaluation is known to depend on option order and label identity (Zheng et al., 2024; Pezeshkpour and Hruschka, 2024), with debiasing usually applied at inference by permuting and averaging. We additionally train against it, which is what makes single-pass serving — the reason this model family exists — cheap again.

Calibration. Temperature scaling (Guo et al., 2017) is the standard post-hoc method; we keep its objective and let the temperature depend on two observable prompt statistics.

Ordinal targets. Squared-EMD losses for ordered classes are standard in label-distribution learning (Hou et al., 2017); we apply them to rubric fields whose expected level the serving path already computes.

Parameter-efficient tuning. LoRA (Hu et al., 2022) with rank-stabilised scaling (Kala-jdziewski, 2023) and LoRA+ (Hayou et al., 2024); QLoRA (Dettmers et al., 2023) for the 24 GB configuration.

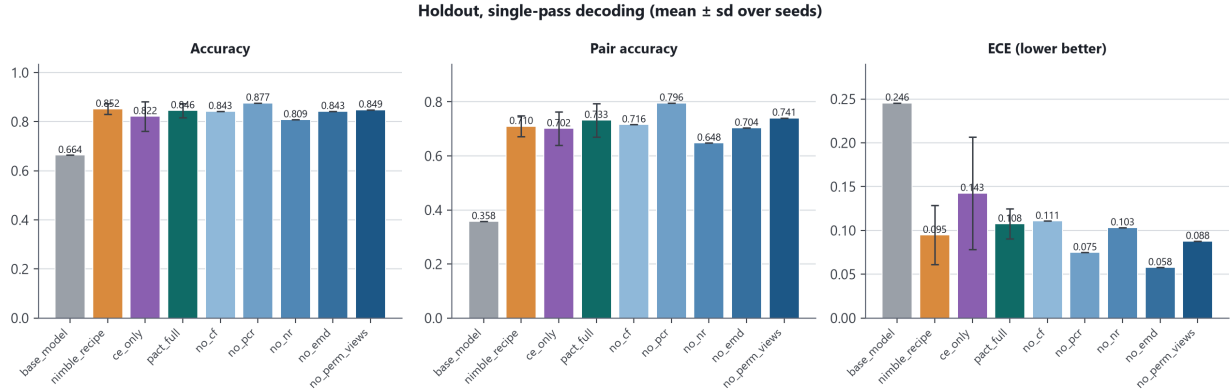


Figure 1: Holdout comparison across runs and ablations, single-pass decoding, mean $\pm$ standard deviation over seeds. Grey: untouched base model. Amber / plum: the two reproduced controls. Teal: the full method. Blue ramp: the five leave-one-out ablations.

8 Limitations

The holdout is 324 synthetic, model-labelled examples from six source families; it is narrow, and its labels were produced by the same family of models that produced the training labels, so shared errors are possible and are not detectable from inside the benchmark. $\mathcal{L}_{NR}$ and $\mathcal{L}_{CF}$ both trust the certificate: if a certificate’s necessity check is wrong, the necessity term teaches indifference where the evidence actually decides. We apply the necessity term to one ablated view per pair, not both, to bound cost. Permutation ensembling multiplies serving cost by K , which is why the headline comparison uses $K = 1$. The five leave-one-out ablations report a single seed each; §6.5’s significance tests are therefore limited to the three multi-seed arms, and a claim about any individual loss term’s effect on raw accuracy (as opposed to pair accuracy or calibration, where the multi-seed arms already carry the evidence) should be read as suggestive, not confirmed. Finally, all claims here concern one base model at one scale with one adapter rank.

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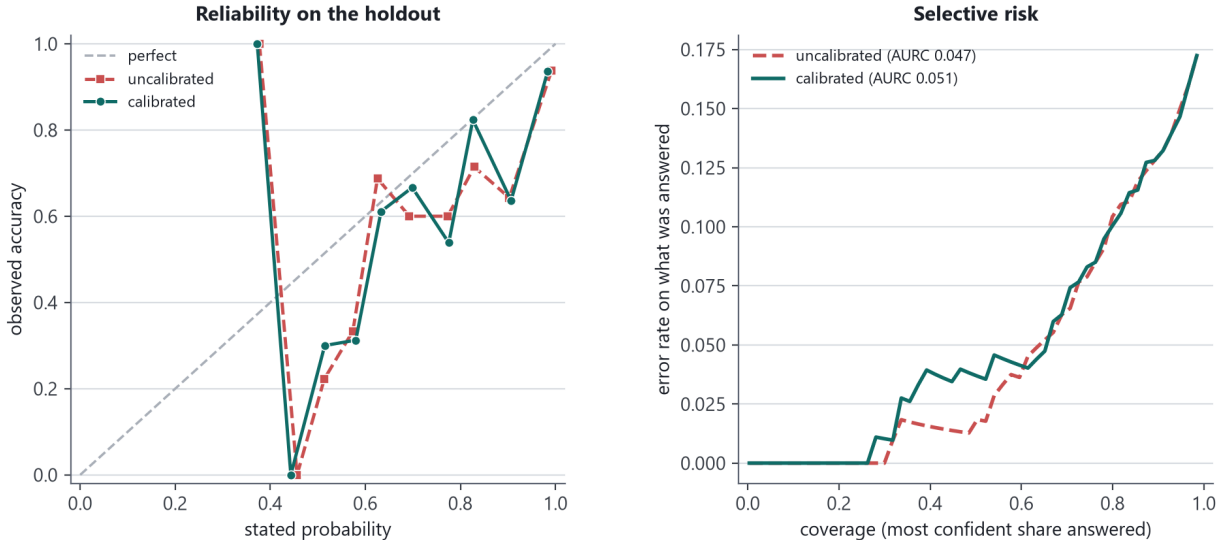


Figure 2: Left: stated probability against observed accuracy on the holdout, before and after the fitted contextual temperature. Right: error rate against the share of examples answered, ranked by confidence.

Multiple-Choice Questions. Findings of NAACL, 2024.

Rogge, N. *Post on single-token decoding for typed decisions*, 2026.

Tang, L., Laban, P., Durrett, G. *MiniCheck: Efficient Fact-Checking of LLMs on Grounding Documents*. EMNLP, 2024.

Zheng, C., Zhou, H., Meng, F., Zhou, J., Huang, M. *Large Language Models Are Not Robust Multiple Choice Selectors*. ICLR, 2024.

A Applied demonstration: an interactive decision task

Everything above evaluates the adapter on the task it was trained for: a static document plus a schema question. As a separate, qualitative check of how far the same serving mechanism generalises, we reused it unmodified for a task with no document at all — choosing where to place falling pieces in a from-scratch Tetris implementation. At every piece, the game engine (not the model) enumerates every legal placement and works out its consequences two plies deep — rows cleared, holes created, resulting stack height and skyline bumpiness, both now and after the best placement of the already-visible next piece. Each option becomes one `choice_description` in an ordinary `PactScorer` schema call, exactly the contract §2.1 describes; the model reads one token and the highest-logit candidate is played. No part of the model, the prompt format or the adapter loading path differs from the main evaluation — only the domain of the schema question changes, from a business document to a game state.

The shipped adapter, unmodified. Given this prompting, the adapter fine-tuned in §5 does not reliably pick the option the two-ply search ranks best, even when that option is marked as clearing a line — the base model this adapter comes from is a general-purpose classifier and was never trained to weigh sequential-decision trade-offs against each other, and PACT’s training objective (§3) does not change that; it targets label accuracy and calibration on the curated pairs,

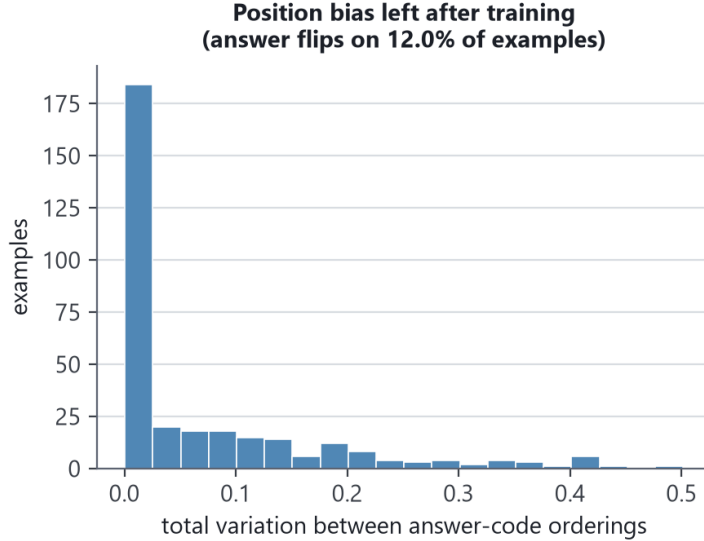


Figure 3: Total-variation distance between the distributions produced under different answer-code orderings.

not action selection. This is the expected outcome, not a failure of §5’s results — it is included here to be precise about what the held-out accuracy numbers do and do not predict.

Closing the gap with policy-gradient fine-tuning. We then further fine-tuned the same LoRA adapter with REINFORCE (single-step, i.e. a contextual bandit over the candidate placements, not full episodic RL), using the exact two-ply heuristic score of the sampled placement as the reward and the mean heuristic score across that state’s candidates as a zero-cost, per-state baseline. No architecture, prompt or answer-code changed — the same **PactScorer**-compatible PEFT adapter that served §5’s evaluation is now trained toward a different objective, with training and serving sharing the identical prompt-construction code so there is no train/serve skew. The fraction of sampled moves matching the heuristic’s own top-ranked candidate (**top_pick_rate**) rose from 0.19 at step 1 to ≥ 0.875 within 200 steps and stayed at 0.75–1.0 for the remainder of 1,574 steps on 2×40GB GPUs (full logs and the trained adapter are in the accompanying repository’s RL/ bundle). Under a fixed 400-piece evaluation cap, greedy (argmax) play scored a mean of 17,940 at step 50 and 17,340 at step 1,550 across 5 games each — flat, because the cap is reached before the model tops out, not because it stopped improving; one uncapped evaluation game at the final checkpoint reached 1,500 pieces, 598 lines and a score of 65,700 before we stopped it.

Reading this appendix. This is a single training run with no seeds, no held-out repetitions and no statistical test, reported for exactly what it is: a demonstration that (a) the PACT/Nimble serving contract is a general typed-choice interface, not something specific to the document benchmark, and (b) a model shown to be a weak sequential decision-maker out of the box under that interface can be measurably improved at the decision task itself, with the same lightweight adapter mechanism, without touching its document-classification behaviour. It should not be read as evidence about §5’s holdout claims in either direction.

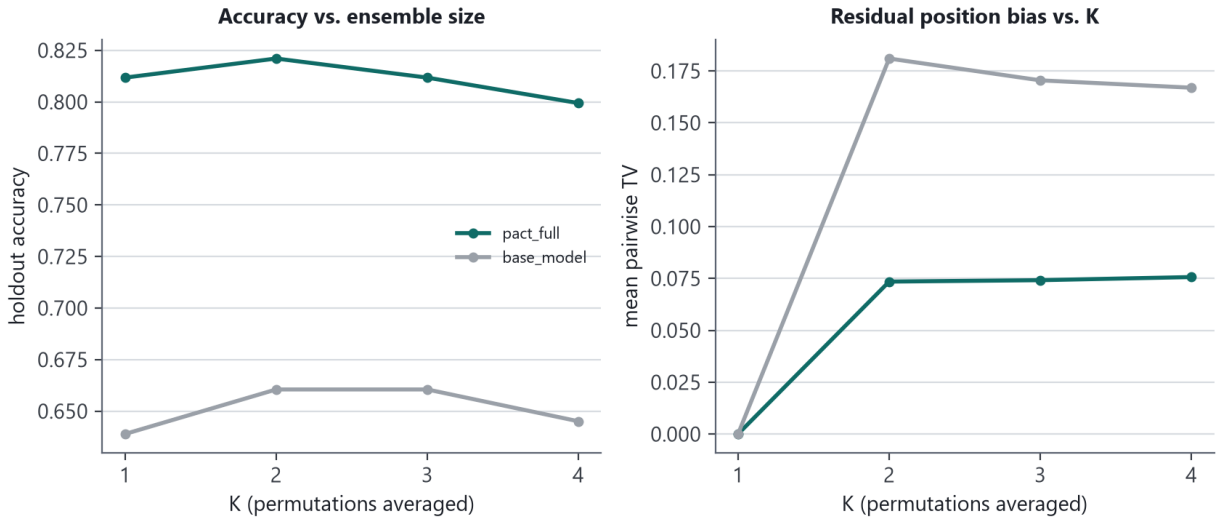


Figure 4: Accuracy and residual position bias as the permutation ensemble size K grows, shipped model vs. base model.